

	GPT-4 Exposure Rating 1		GPT-4 Exposure Rating 2		Human Exposure Rating	
	(1)	(2)	(3)	(4)	(5)	(6)
Software (Webb)	0.00113*** (0.00031)	0.00123*** (0.00031)	0.00111*** (0.00031)	0.00119*** (0.00031)	0.00096*** (0.00031)	0.00101*** (0.00031)
Robot (Webb)	-0.00378*** (0.00032)	-0.00405*** (0.00031)	-0.00377*** (0.00034)	-0.00399*** (0.00033)	-0.00371*** (0.00029)	-0.00383*** (0.00028)
AI (Webb)	0.00080*** (0.00030)	0.00090*** (0.00029)	0.00036 (0.00030)	0.00045 (0.00030)	0.00067** (0.00030)	0.00071** (0.00030)
Suitability for Machine Learning	0.29522*** (0.04503)	0.26888*** (0.04418)	0.28468*** (0.04404)	0.26245*** (0.04342)	0.19514*** (0.03990)	0.18373*** (0.03886)
Normalized Routine Cognitive	0.06601*** (0.00886)	0.06868*** (0.00894)	0.04743*** (0.00872)	0.05015*** (0.00879)	0.03568*** (0.00671)	0.03659*** (0.00669)
Normalized Routine Manual	-0.11147*** (0.00785)	-0.11371*** (0.00789)	-0.09390*** (0.00817)	-0.09561*** (0.00818)	-0.11045*** (0.00741)	-0.11152*** (0.00744)
AI Occupational Exposure Score	0.00993 (0.01107)	0.02465** (0.01059)	-0.01537 (0.01160)	-0.00265 (0.01114)	0.00630 (0.00918)	0.01252 (0.00845)
Frey & Osborne Automation	-0.03024* (0.01835)	-0.03950** (0.01841)	-0.00364 (0.02007)	-0.01217 (0.01972)	-0.03890** (0.01883)	-0.04253** (0.01858)
Log Avg. Salary	0.05804*** (0.01870)		0.04863*** (0.01860)		0.02531 (0.01727)	
Constant	-1.12937*** (0.26859)	-0.45743*** (0.15327)	-0.96117*** (0.26365)	-0.39935*** (0.15017)	-0.47078* (0.24684)	-0.17706 (0.13256)
N	680.00000	681.00000	680.00000	681.00000	680.00000	681.00000
R ²	0.68741	0.68212	0.60737	0.60198	0.71213	0.71126

Table 9: Regression of LLM-exposure scores on prior measures of occupational exposure to AI and automation. We also include annualized wages from the BLS-OES survey in May 2021. Each measure is kept in its original scale, with the exception of routine cognitive and routine manual scores from (Acemoglu and Autor, 2011a). Those two scores are standardized to mean zero and variance 1. Generally we find strong positive associations with previous efforts, though large residual variance to still be explained by our new measures. Columns 1 and 2 are based on our main β exposure measure from GPT-4 ratings. Columns 3 and 4 are based on a similar slightly different exposure rubric also rated by GPT-4 for robustness. Columns 5 and 6 reflect human ratings on the same rubric as columns 1 and 2. Occupation-level scores are built using the core/supplemental task weights, assigning supplemental tasks as having half the weight of core tasks.

6 Discussion

6.1 GPTs as a General-Purpose Technology

Earlier in this paper we discuss the possibility that LLMs could be classified as a general-purpose technology. This classification requires LLMs to meet three core criteria: improvement over time, pervasiveness throughout the economy, and the ability to spawn complementary innovations (Lipsey et al., 2005). Evidence from the AI and machine learning literature thoroughly demonstrates that LLMs meet the first criteria – they are improving in capabilities over time with the ability to complete or be helpful for an increasingly complex set of tasks and use-cases (see 2.1). This paper presents evidence to support the latter two criteria, finding that LLMs on their own can have pervasive impacts across the economy, and that complementary innovations enabled by LLMs – particularly via software and digital tools – can have widespread application to economic activity.

Figure 3 offers one illustration of the potential economic impact of complementary software built on top of LLMs. Taking the difference in the y-axis (the share of all occupations) between α and ζ at a given point along the x-axis (the share of tasks within an occupation that are exposed) gives the aggregate within-occupation exposure potential attributable to tools and software over and above direct exposure from LLMs on their own. The difference in means across all tasks between α and ζ of 0.42 using the GPT-4 annotations and 0.32 using the human annotations (see Figure 3), suggests that the average impact of LLM-powered software on task-exposure may be more than twice as large as the mean exposure from LLMs on their own (mean ζ of 0.14 based on both human annotations and GPT-4 annotations). While our findings suggest that out-of-the-box these models are relevant to a meaningful share of workers and tasks, they also suggest that the software innovations they spawn could drive a much broader impact.

One component of the pervasiveness of a technology is its level of adoption by businesses and users. This paper does not systematically analyze adoption of these models, however, there is early qualitative evidence that adoption and use of LLMs is becoming increasingly widespread. The power of relatively simple UI improvements on top of LLMs was evident in the rollout of ChatGPT – wherein versions of the underlying language model had been previously available via API, but usage skyrocketed after the release of the ChatGPT interface. (Chow, 2023; OpenAI, 2022) Following this release, a number of commercial surveys indicate that firm and worker adoption of LLMs has increased over the past several months. (Constantz, 2023; ResumeBuilder.com, 2023)

Widespread adoption of these models requires addressing existing bottlenecks. A key determinant of their utility is the level of confidence humans place in them and how humans adapt their habits. For instance, in the legal profession, the models’ usefulness depends on whether legal professionals can trust model outputs without verifying original documents or conducting independent research. The cost and flexibility of the technology, worker and firm preferences, and incentives also significantly influence the adoption of tools built on top of LLMs. In this way, adoption may be driven by progress on some of the ethical and safety risks associated with LLMs: bias, fabrication of facts, and misalignment, to name a few OpenAI (2023a). Moreover, the adoption of LLMs will vary across different economic sectors due to factors such as data availability, regulatory environment, and the distribution of power and interests. Consequently, a comprehensive understanding of the adoption and use of LLMs by workers and firms requires a more in-depth exploration of these intricacies.

One possibility is that time savings and seamless application will hold greater importance than quality improvement for the majority of tasks. Another is that the initial focus will be on augmentation, followed by automation (Huang and Rust, 2018). One way this might take shape is through an augmentation phase where jobs first become more precarious (e.g., writers becoming freelancers) before transitioning to full automation.

6.2 Implications for US Public Policy

The introduction of automation technologies, including LLMs, has previously been linked to heightened economic disparity and labor disruption, which may give rise to adverse downstream effects (Acemoglu and Restrepo, 2022a; Acemoglu, 2002; Moll et al., 2021; Klinova and Korinek, 2021; Weidinger et al., 2021, 2022). Our results examining worker exposure in the United States underscore the need for societal and policy preparedness to the potential economic disruption posed by LLMs and the complementary technologies that they spawn. While it is outside the scope of this paper to recommend specific policy prescriptions to smooth the transition to an economy with increasingly widespread LLM adoption, prior work such as (Autor et al., 2022b) has articulated several important directions for US policy related to education, worker training, reforms to safety net programs, and more.

6.3 Limitations and Future Work

In addition to those discussed above, we highlight some particular limitations of this work that warrant further investigation. Primarily, our focus on the United States restricts the generalizability of our findings to other nations where the adoption and impact of generative models may differ due to factors such as industrial organization, technological infrastructure, regulatory frameworks, linguistic diversity, and cultural contexts. We hope to address this limitation by extending the study’s scope and by sharing our methods so other researchers can build on them.

Subsequent research efforts should consider two additional studies: one exploring LLM adoption patterns across various sectors and occupations, and another scrutinizing the actual capabilities and limitations of state-of-the-art models in relation to worker activities beyond the scope of our exposure scores. For example, despite recent advances in multimodal capabilities with GPT-4, we did not consider vision capabilities in the α ratings on direct LLMs-exposure (OpenAI, 2023b). Future work should consider the impact of such capability advances as they unfold. Furthermore, we acknowledge that there may be discrepancies between theoretical and practical performance, particularly in complex, open-ended, and domain-specific tasks.

7 Conclusion

In conclusion, this study offers an examination of the potential impact of LLMs on various occupations and industries within the U.S. economy. By applying a new rubric for understanding LLM capabilities and their potential effects on jobs, we have observed that most occupations exhibit some degree of exposure to LLMs, with higher-wage occupations generally presenting more tasks with high exposure. Our analysis indicates that approximately 19% of jobs have at least 50% of their tasks exposed to LLMs when considering both current model capabilities and anticipated LLM-powered software.

Our research aims to highlight the general-purpose potential of LLMs and their possible implications for US workers. Previous literature demonstrates the impressive improvements of LLMs to date (see 2.1). Our findings confirm the hypothesis that these technologies can have pervasive impacts across a wide swath of occupations in the US, and that additional advancements supported by LLMs, mainly through software and digital tools, can have significant effects on a range of economic activities. However, while the technical capacity for LLMs to make human labor more efficient appears evident, it is important to recognize that social, economic, regulatory, and other factors will influence actual labor productivity outcomes. As capabilities continue to evolve, the impact of LLMs on the economy will likely persist and increase, posing challenges for policymakers in predicting and regulating their trajectory.

Further research is necessary to explore the broader implications of LLM advancements, including their potential to augment or displace human labor, their impact on job quality, impacts on inequality, skill development, and numerous other outcomes. By seeking to understand the capabilities and potential effects

of LLMs on the workforce, policymakers and stakeholders can make more informed decisions to navigate the complex landscape of AI and its role in shaping the future of work.

7.1 LLM Conclusion (GPT-4's Version)

Generative Pre-trained Transformers (GPTs) generate profound transformations, garnering potential technological growth, permeating tasks, greatly impacting professions. This study probes GPTs' potential trajectories, presenting a groundbreaking rubric to gauge tasks' GPT exposure, particularly in the U.S. labor market.

7.2 LLM Conclusion (Author-Augmented Version)

Generative Pre-trained Transformers (GPTs) generate profound transformations, garnering potential technological growth, permeating tasks, gutting professional management. Gauging possible trajectories? Generate pioneering taxonomies, gather policymakers together, generalize past today.

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LLM assistance statement

GPT-4 and ChatGPT were used for writing, coding, and formatting assistance in this project.

A Rubric

A.1 Exposure

E Exposure Taxonomy

Consider the most powerful OpenAI large language model (LLM) This model can complete many tasks that can be formulated as having text input and text output where the context for the input can be captured in 2000 words. The model also cannot draw up-to-date facts (those from <1 year ago) unless they are captured in the input.

Assume you are a worker with an average level of expertise in your role trying to complete the given task. You have access to the LLM as well as any other existing software or computer hardware tools mentioned in the task. You also have access to any commonly available technical tools accessible via a laptop (e.g. a microphone, speakers, etc.). You do not have access to any other physical tools or materials.

Please label the given task according to the taxonomy below.

E0 – No exposure